Descriptions of the required Engineering, mathematics, and science courses are given below.

C. Engineering Courses

Descriptions of the Engineering required courses are given below.

ELEC 210	Electric Circuits	3(3,0)
Circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, RL, and RLC circuits; Laplace Transform; sinusoids and phasors; and an introduction to network theory. Sinusoidal (AC) steady-state, mutual inductance, transformers, and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent.		

ENGR 510	Engineering Project Management	3(3,0)
The course covers key components of engineering project management including projects election and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics and professional conduct, realizing impact of engineering solutions in various contexts (global, economic, environmental, societal, etc...), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

D. Mathematics

A description of the required mathematics courses is given below.

MATH 210	Discrete Mathematics	3(3,0)
Logic, propositional equivalences, predicates and quantifiers, methods of proof, proof strategy, mathematical induction, recursive definitions, strong induction, and structural induction, sets and set operations, functions, growth of functions, relations, and their properties, representing relations, equivalence relations, introduction to graphs, and graph terminology.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.		

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and		

spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.

MATH 311	**Linear Algebra with Applications**	**3(3,0)**
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	**Ordinary Differential Equations**	**3(3,0)**
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 317	**Partial Differential Equations**	**3(3,0)**
Lagrange's theorem, boundary conditions for first-order equations, non-linear first-order PDEs, Charpit's equations, second-order PDEs, classification: hyperbolic, parabolic, and elliptic, the method of separation of variables, introduction to Fourier series and integrals, boundary value problems: heat equation, wave equation, Laplace equation, and finite-length strings. **Prerequisite**: MATH 314.		

MATH 421	**Numerical Analysis**	**3(3,0)**
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

E. Science Courses

A description of the required Sciences courses is given below.

BIOL 210	**Human Anatomy and Physiology and Lab**	**3(2,2)**
A general overview of cell structure and function, and the anatomy and physiology of human body systems. These include cardiovascular, central nervous, respiratory, urinary, digestive, immune, and musculoskeletal systems. The course also offers a set of experiments that deal with basic biological processes and advanced biological assessments. **Prerequisite**: None.		

CHEM 211	**Environmental Chemistry and Lab**	**3(2,2)**
This course provides an opportunity to develop an understanding of several basic environmental functions, the complicated nature of environmental systems, and the need for multidisciplinary solutions to environmental problems. Topics covered include the hydrosphere, water quantity and quality, soil and the soil ecosystem, biological resources, waste disposal, air pollution, ozone depletion, acid rain, and global		

warming. The course also includes a set of experiments that provide students with practical experience in various environmental analysis settings, including air and water quality. **Prerequisite**: None.

GEOL 221	Fundamentals of Geology	3(3,0)
Physical and chemical properties of the earth, structure of the earth, plate tectonic theory and continental drift, volcanism, mountain building processes, surface processes, mineralogy, formation and classification of rocks, earth hazards, geological problems related to civil engineering such as landslides and earthquakes, geologic maps, geophysical exploration methods, earthworks and tunneling. **Prerequisite**: None. Annually.		

PHYS 210	Fundamentals of Physics and Lab	3(3,1)
Welcome to the Introductory Physics course. This course consists of: Review of Classical Mechanics; Fluid Mechanics; General Properties of Waves; Electric Forces; Electric Fields for Discrete and Continuous Charge Distributions; Gauss's Law; Electric Potential; Kirchhoff's Rules; Magnetic Fields and Forces; and Qualitative Discussion of Maxwell's Equations. **Prerequisite**: None.		

PHYS 211	Physics: Electricity and Magnetism and Lab	3(3,1)
This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.		

PHYS 312	Modern Physics	3(3,0)
This course introduces the principles of revolutionary developments of the 20th century. It covers the interaction of light and matter (Photoelectric effect, Compton effect, Auger effect, etc.), the dual nature of light, various models of atomic description, quantum numbers, the relativistic approach, the Heisenberg Uncertainty Principle, the Schrödinger Equation, and an introduction to the band theory of solids and particle physics.		

F. Independent Study Courses

The College of Engineering offers an Independent Study course option at the undergraduate level. Typically, a maximum of two independent studies (totaling not more than 6 credits) may be counted towards your degree as part of your technical elective credits. To begin an Independent Study, you must reach an agreement with a faculty member willing to supervise your work. The supervisor must either be a full-time College of Engineering faculty member or be affiliated with one of the college's engineering departments. The faculty member approves the specific number of credits the proposed work will qualify for, approves a reading list, and meets with you periodically to discuss the reading and your overall progress. 'Independent Study' courses are created by students who work with a faculty member to explore a topic of mutual interest. Students

and faculty in Engineering use independent study courses for several purposes, including a capstone design and/or experimental project, conducting research, or writing a substantial research paper.

Registering for an Independent Study

Follow this procedure to register for an independent study:

1. Identify a faculty advisor and work with them to decide on the content and structure for your course, including how to assess your work, and set up the work timeframe.
2. Complete the “independent study proposal form”.
3. Once your form is submitted, your faculty advisor will review the application. They will either issue approval and forward it to the corresponding department chair or ask you to revise and resubmit your application. You will be notified by email once you can register for the course, after the college issues final approval.

ENGR 597	**Independent Study I**	**3(3,0)**
This course provides students with the possibility to explore new academic research/study that may not be available in regular courses. The student needs to identify a faculty advisor within their department and work with them to decide on the content and structure of the work, including how to assess the student’s work and set the timeframe. The student needs to complete the “independent study proposal form,” which includes an abstract that identifies the nature of the research, goals, and expected outcomes, and submit it to the faculty advisor. Once the form is submitted, the faculty advisor will review the application. They will either issue approval and forward it to the corresponding department chair, ask the student to revise and resubmit the proposal form, or deny it. The student will be notified by email after the college issues final approval. The faculty member approves the specific number of credits the proposed work will qualify for, approves a reading list, and meets with the student periodically to discuss the reading and the overall progress. At the end of the semester, the student is required to submit a paper or a project report acceptable to the faculty supervisor. This course is subject to the same rules and regulations for dropping and withdrawing as any other regular course. **Prerequisite**: Senior Standing		

ENGR 598	**Independent Study II**	**3(3,0)**
This course provides students with the possibility to explore new academic research/study that may not be available in regular courses. The student needs to identify a faculty advisor within their department and work with them to decide on the content and structure of the work, including how to assess it and set the timeframe. The student needs to complete the “independent study proposal form,” which includes an abstract that identifies the nature of the research, goals, and expected outcomes, and submit it to the faculty advisor. Once the form is submitted, the faculty advisor will review the application. They will either issue approval and forward it to the corresponding department chair, ask the student to revise and resubmit the proposal form, or deny it. The student will be notified by email after the college issues final approval. The faculty member approves the specific number of credits the proposed		

work will qualify for, approves a reading list, and meets with the student periodically to discuss the reading and the overall progress. At the end of the semester, the student is required to submit a paper or a project report acceptable to the faculty supervisor. This course is subject to the same rules and regulations for dropping and withdrawing as any other regular course. **Prerequisite**: Senior Standing.

III. Program Requirements

Requirements for the Bachelor of Science and Bachelor of Engineering degrees are program-specific. They encompass two categories: Major and non-major. Each category consists of mandatory and elective courses. The program requirements for the Bachelor of Science and Bachelor of Engineering degrees across the different engineering majors are presented in the following sections. Details and titles of relevant courses are included in the Student's Study Plan (SSP) that every engineering student will have.

Course Coding

The courses offered by the College of Engineering programs are designated according to the following coding System (xxxx abc):

xxxx	Program as indicated above
abc	"a" designates the Year or level 1, 2, 3, 4; "b" program focus area; "c" course sequence - 0, 1, …, 9

The designation used to represent the credit-hour breakdown (t, p) of a course is as follows: "c" is the total credit hours, "t" is the theoretical component of the course, and "p" is the practical or laboratory component. For example, 3(3, 0) denotes a 3-credit-hour course with 3 contact lecture hours and 0 laboratory hours.

Learning Support Center

The Learning Support Center (LSC) at the College of Engineering is another manifestation of RHU's resolute commitment to students' academic success. The LSC offers the students a peer-to-peer learning opportunity that complements classroom learning. The Center aims to support students in advancing their knowledge and skills, reinforcing what they learn in the classroom or overcoming learning and performance difficulties. In addition to being a focal point for inquiry and two-way "asker-replier" learning, the LSC provides space for academic interactions that can significantly enhance learning.

The Center is in room D205 and is open to all students throughout the week. The LSC is staffed by Teaching Assistants, who are graduate and upper-level undergraduate students, and are always available and ready to assist their fellow students in overcoming their academic challenges. Additionally, "on-call" faculty members are assigned to ensure that the LSC is acting in the best interests of the students.

The center's resource portfolio includes books, study guides, course materials, boards, computers, videos, DVDs, documentaries, large-screen LCD panels, and internet-connected PCs.

My Idea to Innovate

My Idea to Innovate (ENGR 499 – 3 credits) is an open course that encourages innovation. A student or a group of students who have a passion for and a dream to nurture and pursue an innovative idea of their own may collaborate with faculty mentors to design the course to cover topics and experiences that could not be achieved within the course offering framework of the concerned program and require collaboration across disciplines and/or fields. Course contents, prerequisites, number of credits, scope, experiences, required resources, outcomes, assessment tools and rubrics, and timeline for completion will be formulated by the student (s) and the faculty mentor(s) once the University Academic Board approves the idea. Student(s) supported by faculty advisor(s) may request approval to design and offer the course after the first year of studies at RHU. The course could count as technical elective credits.